

ECON 101—Introduction to Microeconomics
Fall 2019

Midterm I-Retake

October 11, 2019

Last Name:

First Name:

NU ID number:

Instructions:

- Answer all questions.
- Fill in the answers on this page.
- For the open questions, you must record **only** your final solution here. However, you should provide the procedure for arriving to your final solution in the space provided. Answers without procedure will not be considered.

Multiple choice

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20

Open problems

Problem 1:

1. The coordinate of point B is _____ and the coordinate of point C is _____.
2. The Income effect is _____ and the substitution effect is _____.
3. The cross price elasticity is _____.

Problem 2:

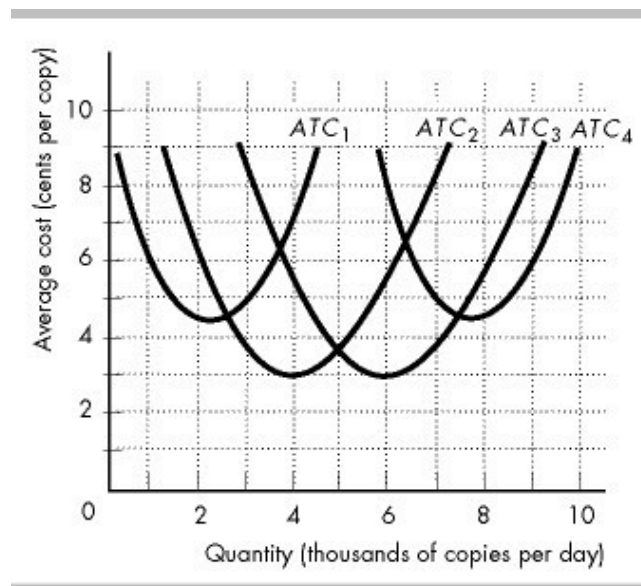
1. The aggregate demand is: _____.
2. The Firm's supply schedule is: _____.
3. The equilibrium price is _____ and the equilibrium quantity is _____.
4. Firms will _____.
5. The long run equilibrium price is _____, the long run equilibrium quantity is _____ and the number of firms is _____.

Part I: Multiple Choice

Please record all questions in the cover sheet. Each question is worth 2 points for a total of 40 points.

1. The marginal cost (MC) curve intersects the
 - (a) ATC and AFC curves at their minimum points.
 - (b) ATC, AVC, and AFC curves at their minimum points.
 - (c) AVC and AFC curves at their minimum points.
 - (d) ATC and AVC curves at their minimum points.
2. Ham and eggs are complements. If the price of ham rises, the demand for eggs will
 - (a) not change but there will be a movement along the demand curve for eggs.
 - (b) decrease and the demand curve for ham will shift rightward.
 - (c) decrease and the demand curve for eggs will shift leftward.
 - (d) increase or decrease but the demand curve for ham will not change.
3. Which of the following is NOT one of the factors that influences the supply of a product?
 - (a) technology
 - (b) income
 - (c) number of suppliers
 - (d) expected future prices
4. The “income effect” in the market for aspirin means that
 - (a) an increase in the price of aspirin will reduce the total purchasing power of aspirin takers, making them able to afford fewer aspirin.
 - (b) a decrease in the price of a substitute good like acetaminophen will make aspirin takers feel a little poorer than they were before.
 - (c) an increase in the price of aspirin will cause headache sufferers to look for a lower priced remedy.
 - (d) aspirin are generally taken by people with higher than average incomes.
5. When the price of a normal good decreases, people buy _____ of the good due to _____.
 - (a) less; the income effect only
 - (b) more; the substitution effect only
 - (c) less; the substitution effect only
 - (d) more; the substitution and income effects
6. If the money price of wheat increases and no other prices change, the
 - (a) opportunity cost of wheat rises.
 - (b) demand for wheat increases.
 - (c) relative price of wheat falls.
 - (d) relative price of wheat is unaffected.
7. In 2014, the price of peanuts was rising, which lead peanut butter sellers and peanut butter buyers to expect the price of peanut butter would rise in the future. In the current market for peanut butter, the price rises and the quantity increases. This set of results means that
 - (a) supply decreased by more than demand increased.
 - (b) demand increased by more than supply increased.
 - (c) supply decreased by more than demand decreased.
 - (d) demand increased by more than supply decreased.
8. Taco Bell’s economists determine that the price elasticity of demand for their tacos is 2.0. So, if Taco Bell raises the price of its tacos by 6.0 percent, the quantity demanded will decrease by _____ percent.
 - (a) 12.0
 - (b) 2.0
 - (c) 6.0
 - (d) 3.0

Consider the following diagram



9. Dustin's copy shop can use four alternative plants. The figure above shows the average total cost curves for Plant 1 (ATC_1), Plant 2 (ATC_2), Plant 3 (ATC_3), and Plant 4 (ATC_4). What is Dustin's long-run average cost if the output is 3,000 copies per day?
- (a) 3.7 cents per copy
 - (b) More information is needed to determine the long-run average cost.
 - (c) 8.5 cents per copy
 - (d) 5.0 cents per copy
10. Each point on the demand curve reflects
- (a) all the wants of a given household.
 - (b) the highest price consumers are willing and able to pay for that particular unit of a good.
 - (c) the highest price sellers will accept for all units they are producing.
 - (d) the lowest-cost technology available to produce a good.

The following table gives the costs at Jan's Bike shop

Labor (workers)	Output (bikes)	Total fixed costs (dollars)	Total variable cost (dollars)	Total cost (dollars)
0	0	200		
1	20		100	
2	50			
3	60			
4	64			

11. The table above gives costs at Jan's Bike Shop. Unfortunately, Jan's record keeping has been spotty. Each worker is paid \$100 a day. Labor costs are the only variable costs of production. What is the total cost of producing 50 bikes?
- (a) \$300
 - (b) \$400
 - (c) \$200
 - (d) \$100

The following table shows the marginal benefit that Marcus derives by consuming different quantities of pizza and pepsi.

Quantity	Pizza	Pepsi
	Marginal Benefit (\$)	Marginal Benefit (\$)
0	0	0
1	20	30
2	16	8
3	10	7

12. The above table shows Priscilla's marginal utility from the two goods she consumes, pizza and Pepsi. Initially, a slice of pizza costs \$4 and a can of Pepsi \$2. Suppose Priscilla has \$12 to spend. If the price of pizza falls to \$2, Priscilla will choose to
- only consume less pizza.
 - consume more Pepsi and more pizza
 - only consume more pizza.
 - only consume more Pepsi.
13. Flights to Paris are a normal good and people's incomes rise. At the same time, the price of jet fuel rises. The equilibrium price of a flight to Paris _____ and the equilibrium quantity of flights to Paris _____.
- rises; increases
 - falls; decreases.
 - rises; might increase, decrease, or not change
 - might rise, fall, or not change; increases
14. Which of the following shifts the supply curve for gasoline rightward?
- an increase in the demand for gas-guzzling, sport utility vehicles
 - an increase in the price of gasoline
 - a decrease in the price of a resource used to produce gasoline, such as crude oil
 - a situation in which the quantity demanded exceeds the quantity supplied

The following table shows the marginal benefit that Steve derives by skiing and skating.

Quantity	Skiing	Skating
	Marginal Benefit (\$)	Marginal Benefit (\$)
0		
1	50	70
2	40	40
3	30	30
4	20	20
5	10	5

15. Steve has two goods he can spend his income on, skiing and skating, and his marginal utilities from each are in the table above. The price of each unit of skiing is \$10 and the price of each unit of skating is \$5. Steve has \$40 to spend. What quantities of skiing and skating should Steve consume to maximize his utility?
- 2 units of skiing and 4 units of skating
 - 1 unit of skiing and 2 units of skating
 - 4 units of skiing and 4 units of skating
 - 5 units of skiing and 5 units of skating

16. Suppose that people find out that eating more fish improves their health, leading them to increase their demand for fish. As a result, the equilibrium price of fish _____ and the equilibrium quantity _____.
- (a) rises; increases
 - (b) falls; increases
 - (c) falls; decreases
 - (d) rises; decreases
17. An ice cream cone costs \$1.50. A can of soda costs \$0.75. The relative price of an ice cream cone is
- (a) \$1.50, the opportunity cost of a can of soda.
 - (b) 2 cans of soda per ice cream cone, the opportunity cost of an ice cream cone.
 - (c) 1/2 can of soda per ice cream cone, the opportunity cost of an ice cream cone.
 - (d) \$0.75, the opportunity cost of a can of soda.
18. In March, the quantity of orange juice sold in the town of Jackson was 3000 cartons and the price \$3. In May, the quantity of orange juice sold in the town of Jackson was 3500 cartons and the price was \$3.20. This change in the price and quantity sold could have been the result of
- (a) the after effects of a cold winter in Florida that killed half of the orange crop.
 - (b) the after effects of a warm winter in Florida that increased the orange crop yield by 50 percent.
 - (c) the release of a medical study suggesting that consuming orange juice helps prevent cancer.
 - (d) a reduction in the number of orange juice coupons provided by local markets.
19. "Last October, due to an early frost, the price of a pumpkin increased by 10 percent compared to the price in the previous Halloween seasons. As a result, the quantity demanded county-wide decreased from 2 million to 1.5 million." Based on this statement, the
- (a) demand for pumpkins is inelastic.
 - (b) demand for pumpkins is elastic.
 - (c) demand curve for pumpkins shifted rightward.
 - (d) demand for pumpkins is unit elastic.
20. If the price of a video rental is below its equilibrium price, there will be a _____ of video rentals and the price will _____.
- (a) shortage; rise
 - (b) surplus; fall
 - (c) surplus; rise
 - (d) shortage; fall

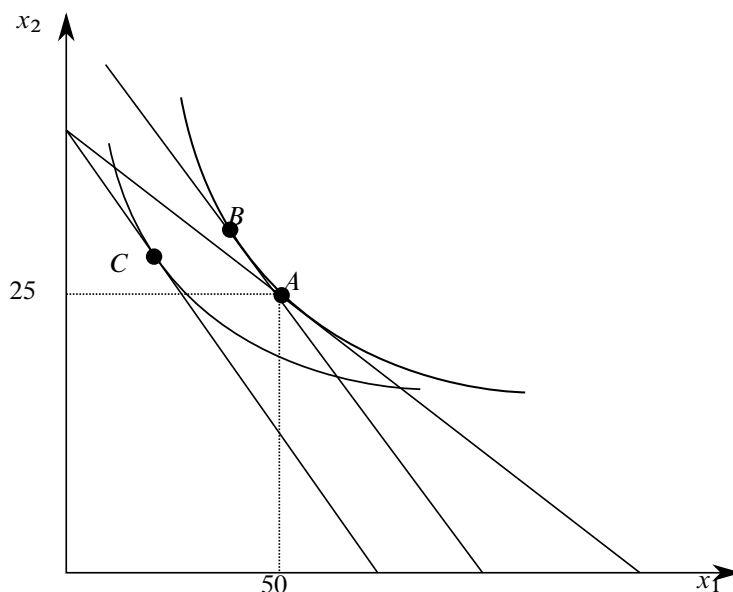
End of Multiple Choice Section: Please don't forget to record your answers in the cover sheet!

Continue to Part 2

Part II: Open problems

Instructions: Please do your calculations here but write your final answers in the space provided in the cover sheet.

1. Consider the following scenario as depicted by the following figure.



This consumer can consume goods x_1 and x_2 . The consumer has an income of \$100 dollars. The price of x_1 is \$1 dollar and the price of x_2 is \$2. Each consumer initially chooses the initial consumption bundle $x_1 = 50$, $x_2 = 25$ which is denoted by point A. Suppose that there is an increase in the price of x_1 to \$2. After this price increase, the optimal consumption bundle of the consumer is denoted by point C. Suppose the government gives the consumer a monetary transfer of \$60 so that she can achieve her original indifference curve before the price change. Also, assume that an economic researcher in the bureau of taxation gives you the following information:

- The arc own price elasticity for good 1 of the consumer is $|\epsilon_{x_1}^{p_1}| = 1.29$.
- The arc income elasticity with respect to good 1 of the consumer is $\epsilon_{x_1}^I = 1.44$

With the previous information please compute (round your answers to the nearest hundredth):

- The coordinates of points B and C.
- The Income and Substitution effects.
- The (arc) cross price elasticity of good x_2

Write your procedure for finding the solution here.

2. Suppose that an economy has two individuals, A and B. A's demand curve is given by $q_D^A = 100,000 - 2p$ and B's demand curve is given by $q_D^B = 150,000 - 6P$. Furthermore, there are initially 10,000 identical firms with the following cost structure: Total Cost: $TC(q) = 10,000 + 100q + 10q^2$, Marginal Cost: $MC(q) = 100 + 20q$.

- (a) Find the market demand curve. Provide an expression for Q_D
- (b) What is the firm's shutdown point and what is the firm's supply schedule?
- (c) What is the market price and quantity?
- (d) Will firms enter or exit?
- (e) Calculate the long-run equilibrium price, quantity, and number of firms.

Write your procedure for finding the solution here. Round your answers to the nearest hundredth.

